

The Basic Trigonometric Identities

Name
Date
Course

Operations with Trigonometric Functions
Simplify.

• ① $3\sin\theta + 5\sin\theta$ ② $37\cot A - 20\cot A$

$\boxed{8\sin\theta}$

$\boxed{17\cot A}$

• ③ $\cos\beta + \cos\beta$ ④ $-9\sec\beta + 2\sec\beta$

$\boxed{2\cos\beta}$

$\boxed{-7\sec\beta}$

$\sin\theta \cdot \sin\theta \neq \sin\theta^2$

rather

$\sin\theta \cdot \sin\theta = (\sin\theta)^2 \text{ or } \sin^2\theta$

• ⑤ $\cos A \cdot \cos A$ ⑥ $\tan^2\beta \cdot \tan^3\beta$ Exponent Rule I

$\boxed{\cos^2 A}$

$\boxed{\tan^5\beta}$

$x^a \cdot x^b = x^{a+b}$

• ⑦ $\csc W \cdot \csc W$ ⑧ $\cot^8 B \cdot \cot^3 B$

$\boxed{\csc^2 W}$

$\boxed{\cot^{11} B}$

Exponent Rule II

• ⑨ $\frac{\sec^{10}\alpha}{\sec^4\alpha} = \boxed{\sec^6\alpha}$ ⑩ $\frac{\sin^5\theta}{\sin^3\theta} = \boxed{\sin^2\theta}$ $\frac{x^a}{x^b} = x^{a-b}$

• ⑪ $\frac{\tan^9 x}{\tan^6 x} = \boxed{\tan^3 x}$ ⑫ $\frac{\csc^3 y}{\csc^2 y} = \boxed{\csc y}$

The Basic Trigonometric Identities

Reciprocal Identities

$\sin\theta = \frac{1}{\csc\theta}$

$\csc\theta = \frac{1}{\sin\theta}$

$\cos\theta = \frac{1}{\sec\theta}$

$\sec\theta = \frac{1}{\cos\theta}$

$\tan\theta = \frac{1}{\cot\theta}$

$\cot\theta = \frac{1}{\tan\theta}$

Ratio Identities

$\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\cot\theta = \frac{\cos\theta}{\sin\theta}$

Pythagorean Identities

$\sin^2\theta + \cos^2\theta = 1$

$\tan^2\theta + 1 = \sec^2\theta$

$1 + \cot^2\theta = \csc^2\theta$

Brief Explanations of the Basic Identities

Reciprocal Identities

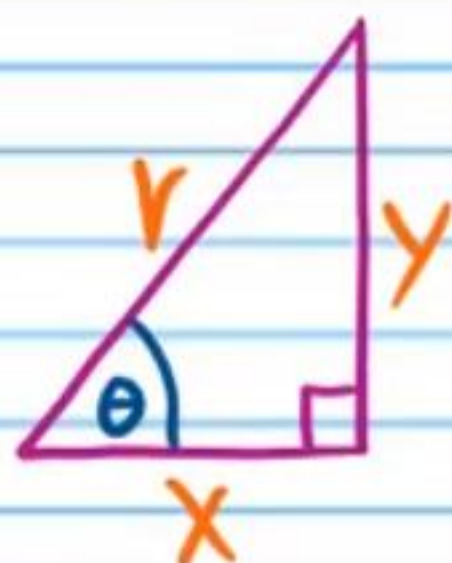
$$\sin \theta = \frac{1}{\csc \theta}$$

$$\frac{y}{r} = \frac{1}{\frac{r}{y}}$$

$$\frac{y}{r} = 1 \div \frac{r}{y}$$

$$\frac{y}{r} = 1 \cdot \frac{y}{r}$$

$$\frac{y}{r} = \frac{y}{r} \quad \checkmark$$



Ratio Identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\frac{y}{x} = \frac{\frac{y}{r}}{\frac{x}{r}}$$

$$\frac{y}{x} = \frac{y}{r} \div \frac{x}{r}$$

$$\frac{y}{x} = \frac{y}{r} \cdot \frac{r}{x}$$

$$\frac{y}{x} = \frac{y}{x} \quad \checkmark$$

Pythagorean Identities

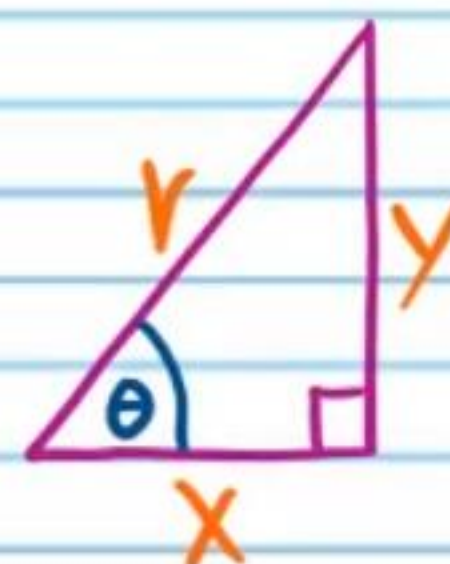
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{y}{r}\right)^2 + \left(\frac{x}{r}\right)^2 = 1$$

$$\frac{y^2}{r^2} + \frac{x^2}{r^2} = 1$$

Exponent Rule IV

$$\left(\frac{m}{n}\right)^a = \frac{m^a}{n^a}$$



$$\frac{y^2 + x^2}{r^2} = 1$$

$$\frac{r^2}{r^2} = 1$$

$$1 = 1 \quad \checkmark$$

Simplifying Trigonometric Expressions

Using the Reciprocal Identities to Simplify Expressions
Simplify.

• (13) $\sin \theta \csc \theta$

$$\sin \theta \frac{1}{\sin \theta}$$

$$\frac{\sin \theta}{\sin \theta}$$

$$\boxed{1}$$

• (15) $1 + \cos \theta \sec \theta$

$$1 + \cos \theta \frac{1}{\cos \theta}$$

$$1 + 1$$

$$\boxed{2}$$

(14) $\cot w \tan w - 1$

$$\frac{1}{\tan w} \tan w - 1$$

$$\frac{\tan w}{\tan w} - 1$$

$$1 - 1$$

$$\boxed{0}$$

(16) $2 - \tan \beta \cot \beta$

$$2 - \tan \beta \frac{1}{\tan \beta}$$

$$2 - 1$$

$$\boxed{1}$$

• (17) $2 \cos A + \csc A \cos A \sin A$ (18) $\cot x \sec x \tan x - \sec x$

$$2 \cos A + \frac{1}{\sin A} \cos A \sin A$$

$$2 \cos A + \cos A$$

$$\boxed{3 \cos A}$$

$$\frac{1}{\tan x} \sec x \tan x - \sec x$$

$$\sec x - \sec x$$

$$\boxed{0}$$

• (19) $\cos B \sin B \sec B - 3 \sin B$ (20) $5 \csc w + \tan w \csc w \cot w$

$$\cos B \sin B \frac{1}{\cos B} - 3 \sin B$$

$$\sin B - 3 \sin B$$

$$\boxed{-2 \sin B}$$

$$5 \csc w + \tan w \csc w \frac{1}{\tan w}$$

$$5 \csc w + \csc w$$

$$\boxed{6 \csc w}$$

• (21) $\frac{1}{\tan \beta} + \frac{1}{\tan \beta}$

$$\cot \beta + \cot \beta$$

$$\boxed{2 \cot \beta}$$

(22) $\frac{3}{\cos \theta} - \frac{2}{\cos \theta}$

$$3 \left(\frac{1}{\cos \theta} \right) - 2 \left(\frac{1}{\cos \theta} \right)$$

$$3 \sec \theta - 2 \sec \theta$$

$$\boxed{\sec \theta}$$

$$\bullet \textcircled{23} \frac{1}{\sin W} + \frac{1}{\sin W} + \frac{1}{\sin W} \quad \textcircled{24} \frac{5}{\cos X} + \frac{8}{\cos X} - \frac{2}{\cos X}$$

$$\csc W + \csc W + \csc W \quad 5 \sec X + 8 \sec X - 2 \sec X$$

$$\boxed{3 \csc W}$$

$$\boxed{11 \sec X}$$

$$\bullet \textcircled{25} \frac{1}{\cos^2 A}$$

$$\textcircled{26} \frac{1}{\csc^2 B}$$

$$\frac{1}{\cos A} \cdot \frac{1}{\cos A}$$

$$\boxed{\sin^2 B}$$

$$\sec A \cdot \sec A$$

$$\boxed{\sec^2 A}$$

$$\bullet \textcircled{27} \frac{1}{\tan^2 \theta}$$

$$\textcircled{28} \frac{1}{\sec^2 \theta}$$

$$\boxed{\cot^2 \theta}$$

$$\boxed{\cos^2 \theta}$$

Using the Ratio Identities to Simplify Expressions
Simplify.

$$\bullet \textcircled{29} \cos \theta \tan \theta$$

$$\textcircled{30} \frac{\cos \theta}{\cot \theta}$$

$$\cos \theta \frac{\sin \theta}{\cos \theta}$$

$$\cos \theta \div \cot \theta$$

$$\frac{\cos \theta \sin \theta}{\cos \theta}$$

$$\cos \theta \div \frac{\cos \theta}{\sin \theta}$$

$$\boxed{\sin \theta}$$

$$\cos \theta \cdot \frac{\sin \theta}{\cos \theta}$$

$$\boxed{\sin \theta}$$

Exponent Rule IV

$$\frac{x^a}{y^a} = \left(\frac{x}{y}\right)^a$$

$$\textcircled{31} \frac{\sin \beta}{\cos \beta} - \tan \beta$$

$$\textcircled{32} \frac{\sin^2 A}{\cos^2 A}$$

$$\tan \beta - \tan \beta$$

$$\left(\frac{\sin A}{\cos A}\right)^2$$

$$\boxed{0}$$

$$\boxed{\tan^2 A}$$

$$\boxed{\tan^2 A}$$

• (33) $\cot \theta \sin \theta$

$$\frac{\cos \theta}{\sin \theta} \sin \theta$$

$$\boxed{\cos \theta}$$

(34) $\frac{\sin \theta}{\tan \theta}$

$$\sin \theta \div \tan \theta$$

$$\sin \theta \div \frac{\sin \theta}{\cos \theta}$$

$$\sin \theta \cdot \frac{\cos \theta}{\sin \theta}$$

$$\boxed{\cos \theta}$$

(35) $\cot w + \frac{\cos w}{\sin w}$

$$\cot w + \cot w$$

$$\boxed{2 \cot w}$$

(36) $\frac{\cos^2 A}{\sin^2 A}$

$$\boxed{\cot^2 A}$$

Using the Pythagorean Identities to Simplify Expressions
Simplify.

• (37) $\sin^2 \theta + \cos^2 \theta - 1$

$$1 - 1$$

$$\boxed{0}$$

(38) $1 - \sin^2 \theta$

$$\sin^2 \theta + \cos^2 \theta - \sin^2 \theta$$

$$\boxed{\cos^2 \theta}$$

(39) $\frac{1 + \cot^2 B}{\csc B}$

$$\frac{\csc^2 B}{\csc B}$$

$$\boxed{\csc B}$$

(40) $-\tan^2 B + \sec^2 B$

$$-\tan^2 B + \tan^2 B + 1$$

$$\boxed{1}$$

• (41) $4 + \sin^2\theta + \cos^2\theta$

$4 + 1$

$\boxed{5}$

(42) $1 - \cos^2\theta$

$\sin^2\theta + \cos^2\theta - \cos^2\theta$

$\boxed{\sin^2\theta}$

(43) $\frac{\tan^2 w + 1}{\sec w}$

$\frac{\sec^2 w}{\sec w}$

$\boxed{\sec w}$

(44) $\csc^2\theta - \cot^2\theta + 1$

$1 + \cot^2\theta - \cot^2\theta + 1$

$1 + 1$

$\boxed{2}$

FOIL

• (45) $(1 + \sin w)(1 - \sin w)$

$1 \cdot 1 + 1 \cdot (-\sin w) + 1 \cdot \sin w + \sin w(-\sin w)$

$1 - \sin w + \sin w - \sin^2 w$

$1 - \sin^2 w$

$\sin^2 w + \cos^2 w - \sin^2 w$

$\boxed{\cos^2 w}$

(46) $(\sec\beta + 1)(\sec\beta - 1) = \sec^2\beta - \sec\beta + \sec\beta - 1$

$= \sec^2\beta - 1$

$= \tan^2\beta + 1 - 1$

$= \boxed{\tan^2\beta}$

• (47) $(\cos\alpha + 1)(1 - \cos\alpha) = \cos\alpha - \cos^2\alpha + 1 - \cos\alpha$

$= -\cos^2\alpha + 1$

$= -\cos^2\alpha + \sin^2\alpha + \cos^2\alpha$

$= \boxed{\sin^2\alpha}$

$$\begin{aligned}
 \bullet (48) (\csc x - 1)(\csc x + 1) &= \csc^2 x - 1 \\
 &= \csc^2 x - 1 \\
 &= 1 + \cot^2 x - 1 \\
 &= \boxed{\cot^2 x}
 \end{aligned}$$

Using the Reciprocal and Ratio Identities to Simplify Expressions

$$\bullet (49) \frac{1}{\frac{\sin \theta}{\cos \theta}}$$

$$\frac{1}{\tan \theta}$$

$$\boxed{\cot \theta}$$

$$(50) \sec B \sin B$$

$$\frac{1}{\cos B} \sin B$$

$$\frac{\sin B}{\cos B}$$

$$\boxed{\tan B}$$

$$(51) \sin \theta \cot \theta \sec \theta - 1$$

$$\cancel{\sin \theta} \frac{\cos \theta}{\cancel{\sin \theta}} \frac{1}{\cancel{\cos \theta}} - 1$$

$$1 - 1$$

$$\boxed{0}$$

$$(52) \tan \theta \csc \theta$$

$$\frac{\cancel{\sin \theta}}{\cos \theta} \frac{1}{\cancel{\sin \theta}}$$

$$\frac{1}{\cos \theta}$$

$$\boxed{\sec \theta}$$

$$\bullet (53) \frac{1}{\frac{\cos A}{\sin A}}$$

$$\frac{1}{\cot A}$$

$$\boxed{\tan A}$$

$$(54) \cos A \csc A$$

$$\cos A \frac{1}{\sin A}$$

$$\frac{\cos A}{\sin A}$$

$$\boxed{\cot A}$$

$$(55) \tan \theta \cos \theta \csc \theta$$

$$\frac{\cancel{\sin \theta}}{\cancel{\cos \theta}} \cancel{\cos \theta} \frac{1}{\cancel{\sin \theta}}$$

1

$$(56) \cot \theta \sec \theta$$

$$\frac{\cancel{\cos \theta}}{\cancel{\sin \theta}} \frac{1}{\cancel{\cos \theta}}$$

$$\frac{1}{\cancel{\sin \theta}}$$

$$\boxed{\csc \theta}$$

$$(57) \frac{\tan \theta}{\sec \theta} = \tan \theta \div \sec \theta = \frac{\sin \theta}{\cos \theta} \div \frac{1}{\cos \theta}$$

$$= \frac{\cancel{\sin \theta}}{\cancel{\cos \theta}} \cdot \frac{\cancel{\cos \theta}}{1}$$

$$= \frac{\cancel{\sin \theta}}{1}$$

$$= \boxed{\sin \theta}$$

$$(58) \frac{\sec y}{\csc y} = \sec y \div \csc y = \frac{1}{\cos y} \div \frac{1}{\sin y}$$

$$= \frac{1}{\cos y} \cdot \frac{\sin y}{1}$$

$$= \frac{\sin y}{\cos y}$$

$$= \boxed{\tan y}$$

$$(59) \frac{\cot \theta}{\csc \theta} = \cot \theta \div \csc \theta = \frac{\cos \theta}{\sin \theta} \div \frac{1}{\sin \theta}$$

$$= \frac{\cancel{\cos \theta}}{\cancel{\sin \theta}} \cdot \frac{\cancel{\sin \theta}}{1}$$

$$= \boxed{\cos \theta}$$

$$\begin{aligned}
 \textcircled{60} \quad \frac{\csc \theta}{\sec \theta} &= \csc \theta \div \sec \theta = \frac{1}{\sin \theta} \div \frac{1}{\cos \theta} \\
 &= \frac{1}{\sin \theta} \cdot \frac{\cos \theta}{1} \\
 &= \frac{\cos \theta}{\sin \theta} \\
 &= \boxed{\cot \theta}
 \end{aligned}$$

Using the Pythagorean and Reciprocal Identities to Simplify Expressions

$$\begin{aligned}
 \textcircled{61} \quad \frac{\sin^2 \theta + \cos^2 \theta}{\sec \theta} &= \frac{1}{\sec \theta} \\
 &= \boxed{\cos \theta}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{62} \quad \frac{1}{\tan^2 \theta + 1} &= \frac{1}{\sec^2 \theta} \\
 &= \boxed{\cos^2 \theta}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{63} \quad \cos^2 \theta (\tan^2 \theta + 1) &= \cos^2 \theta \sec^2 \theta \\
 &= \cos^2 \theta \cdot \frac{1}{\cos^2 \theta} \\
 &= \frac{\cos^2 \theta}{\cos^2 \theta} \\
 &= \boxed{1}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{64} \quad \frac{\sin^2 B + \cos^2 B}{\csc^2 B} &= \frac{1}{\csc^2 B} \\
 &= \boxed{\sin^2 B}
 \end{aligned}$$

$$\bullet \textcircled{65} \frac{\sin^2 \theta + \cos^2 \theta}{\csc \theta}$$

$$\frac{1}{\csc \theta}$$

$$\boxed{\sin \theta}$$

$$\textcircled{67} \sin^2 \beta (1 + \cot^2 \beta)$$

$$\sin^2 \beta \csc^2 \beta$$

$$\sin^2 \beta \frac{1}{\sin^2 \beta}$$

$$\frac{\sin^2 \beta}{\sin^2 \beta}$$

$\boxed{1}$

$$\textcircled{66} \frac{1}{1 + \cot^2 \theta}$$

$$\frac{1}{\csc^2 \theta}$$

$$\boxed{\sin^2 \theta}$$

$$\textcircled{68} \frac{\cos^2 A + \sin^2 A}{\cot^2 A}$$

$$\frac{1}{\cot^2 A}$$

$$\boxed{\tan^2 A}$$

$$\bullet \textcircled{69} \sin^2 \theta + \sin \theta \csc \theta + \cos^2 \theta \quad \textcircled{70} 1 + \cot \theta + \tan^2 \theta + \cot^2 \theta$$

$$\sin^2 \theta + \cos^2 \theta + \sin \theta \csc \theta$$

$$1 + \frac{1}{\csc \theta} \csc^2 \theta$$

$$1 + \frac{\csc^2 \theta}{\csc \theta}$$

$$\boxed{1 + \csc \theta}$$

$$1 + \cot^2 \theta + \cot \theta \tan^2 \theta$$

$$\csc^2 \theta + \frac{1}{\tan \theta} \tan^2 \theta$$

$$\csc^2 \theta + \frac{\tan^2 \theta}{\tan \theta}$$

$$\boxed{\csc^2 \theta + \tan \theta}$$

$$\bullet \textcircled{71} \cos^2 A + 2 + \sin^2 A - \cos^2 A \sec A$$

$$\cos^2 A + \sin^2 A + 2 - \cos^2 A \sec A$$

$$1 + 2 - \cos^2 A \frac{1}{\cos A}$$

$$3 - \frac{\cos^2 A}{\cos A}$$

$$\boxed{3 - \cos A}$$

$$\textcircled{72} \tan^2 W + \sin W + 1 - \sec^3 W \cos W$$

$$\tan^2 W + 1 + \sin W - \sec^3 W \cos W$$

$$\sec^2 W + \sin W - \sec^3 W \frac{1}{\sec W}$$

$$\sec^2 W + \sin W - \frac{\sec^3 W}{\sec W}$$

$$\sec^2 W + \sin W - \sec^2 W$$

$$\boxed{\sin W}$$

Using the Pythagorean and Ratio Identities to Simplify Expressions

$$\bullet \textcircled{73} 1 + \frac{\cos^2 B}{\sin^2 B}$$

$$1 + \cot^2 B$$

$$\boxed{\csc^2 B}$$

$$\bullet \textcircled{75} \frac{\sin^2 A}{\cos^2 A} + 1$$

$$\tan^2 A + 1$$

$$\boxed{\sec^2 A}$$

$$\textcircled{74} \frac{1 - \cos^2 \alpha}{1 - \sin^2 \alpha}$$

$$\frac{\sin^2 \alpha + \cos^2 \alpha - \cos^2 \alpha}{\sin^2 \alpha + \cos^2 \alpha - \sin^2 \alpha}$$

$$\frac{\sin^2 \alpha}{\cos^2 \alpha}$$

$$\tan^2 \alpha$$

$$\textcircled{76} \frac{1 - \sin^2 \theta}{1 - \cos^2 \theta}$$

$$\frac{\sin^2 \theta + \cos^2 \theta - \sin^2 \theta}{\sin^2 \theta + \cos^2 \theta - \cos^2 \theta}$$

$$\frac{\cos^2 \theta}{\sin^2 \theta}$$

$$\boxed{\cot^2 \theta}$$

$$\bullet \textcircled{77} \frac{\sec^2 \beta - 1 + \cos \beta - \tan^2 \beta}{\sin \beta + \cot^2 \beta + 1 - \csc^2 \beta}$$

$$\frac{\tan^2 \beta + 1 - 1 + \cos \beta - \tan^2 \beta}{\sin \beta + \csc^2 \beta - \csc^2 \beta}$$

$$\frac{\cos \beta}{\sin \beta}$$

$$\boxed{\cot \beta}$$

$$\bullet \textcircled{78} \frac{\cos^2 \alpha + \sin^2 \alpha + \sin \alpha - 1}{-1 + \cos \alpha + \sec^2 \alpha - \tan^2 \alpha}$$

$$\frac{1 + \sin \alpha - 1}{-1 + \cos \alpha + \tan^2 \alpha + 1 - \tan^2 \alpha}$$

$$\frac{\sin \alpha}{\cos \alpha}$$

$$\boxed{\tan \alpha}$$

Using All Three Types of Basic Identities to Simplify Expressions

$$\bullet \textcircled{79} -1 + \csc^2 \theta \tan^2 \theta$$

$$-1 + \frac{1}{\sin^2 \theta} \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$-1 + \frac{\sin^2 \theta}{\sin^2 \theta \cos^2 \theta}$$

$$-1 + \frac{1}{\cos^2 \theta}$$

$$-1 + \sec^2 \theta$$

$$-1 + \tan^2 \theta + 1$$

$$\boxed{\tan^2 \theta}$$

$$\textcircled{80} \frac{1 + \cot^2 w}{1 + \tan^2 w}$$

$$\frac{\csc^2 w}{\sec^2 w}$$

$$\csc^2 w \div \sec^2 w$$

$$\frac{1}{\sin^2 w} \div \frac{1}{\cos^2 w}$$

$$\frac{1}{\sin^2 w} \cdot \frac{\cos^2 w}{1}$$

$$\frac{\cos^2 w}{\sin^2 w}$$

$$\boxed{\cot^2 w}$$

• (81) $\sec^2 \theta \cot^2 \theta - 1$

$$\frac{1}{\cos^2 \theta} \frac{\cos^2 \theta}{\sin^2 \theta} - 1$$

$$\frac{\cos^2 \theta}{\cos^2 \theta \sin^2 \theta} - 1$$

$$\frac{1}{\sin^2 \theta} - 1$$

$$\csc^2 \theta - 1$$

$$1 + \cot^2 \theta - 1$$

$$\boxed{\cot^2 \theta}$$

(82) $\frac{\tan^2 x + 1}{1 + \cot^2 x}$

$$\frac{\sec^2 x}{\csc^2 x}$$

$$\sec^2 x \div \csc^2 x$$

$$\frac{1}{\cos^2 x} \div \frac{1}{\sin^2 x}$$

$$\frac{1}{\cos^2 x} \cdot \frac{\sin^2 x}{1}$$

$$\frac{\sin^2 x}{\cos^2 x}$$

$$\boxed{\tan^2 x}$$